higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

DIGITAL ELCTRONICS N4

16 AUGUST 2023

This marking guideline consists of 6 pages.

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QUESTION 1

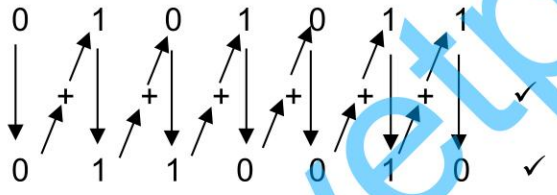
1.1 B
1.2 D
1.3 D
1.4 C
1.5 A
1.6 D

(6 × 1) **[6]**

QUESTION 2

2.1 Self-complementing codes are the codes that have the property that 9's complement of a decimal number✓ is obtained directly by replacing 1s by 0s and 0s by 1s, i.e. by complementing each bit in the pattern.✓ (2)

2.2



∴ 0101011 Gray = 0110010_2✓ (3)

2.3 2.3.1 $6,5_8 = 0110,101_2$✓
$9,75_{10} = 1001,11_2$✓

2's complement of 1001,110 = 0110,001 + 1 = 0110,010✓

0	*	0	1	1	0,	1	0	1	
1	*	0	1	1	0,	0	1	0	
1	*	1	1	0	0,	1	1	1	✓

2's complement of 1*1100,111 = -0011,000 + 1 = -0011,001✓
Answer in hexadecimal system: -0011, $0010_2 = -3.2_{16}$✓ (6)

2.3.2 $4,75_{10} = 100,11_2$✓
$15,5_8 = 001101,101_2$✓

					1	1	0	1,	1	0	1		
					X	1	0	0,	1	1	0		
					0	0	0	0	0	0	0	✓	
				1	1	0	1	1	0	1			
			1	1	0	1	1	0	1				
		0	0	0	0	0	0	0				✓	
	0	0	0	0	0	0	0						
+	1	1	0	1	1	0	1						
1	0	0	0	0	0	0	1	0	1	1	1	0	✓

∴ Answer $1000000,10111_2 = 100,56_8$✓ (6)

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2.3.3 $14{,}625_{10} = 1110{,}101_2$ ✓
$4{,}4_8 = 100{,}100_2$ ✓

$$
\begin{array}{r|l}
 & 1\ 1,\ 0\ 1\ \checkmark \\
\hline
1\ 0\ 0\ 1\ 0\ 0 & 1\ 1\ 1\ 0\ 1\ 0,\ 0\ 0 \\
 & 1\ 0\ 0\ 1\ 0\ 0\ \downarrow\ \downarrow\ \downarrow \\
\hline
 & 1\ 0\ 1\ 1\ 0\ 1\ \downarrow\ \downarrow\ \checkmark \\
 & 1\ 0\ 0\ 1\ 0\ 0\ \downarrow\ \downarrow \\
\hline
 & 0\ 0\ 1\ 0\ 0\ 1\ 0\ \downarrow\ \checkmark \\
 & 0\ 0\ 0\ 0\ 0\ 0\ \downarrow \\
\hline
 & 1\ 0\ 0\ 1\ 0\ 0\ \checkmark \\
 & 1\ 0\ 0\ 1\ 0\ 0 \\
\hline
 & 0\ 0\ 0\ 0\ 0\ 0
\end{array}
$$

(6)
[23]

QUESTION 3

3.1 3.1.1 A
3.1.2 1
3.1.3 A + B

(3 × 1) (3)

3.2 3.2.1 $\bar{A}BC + A\bar{B}C + \bar{A}\bar{B}C + \bar{A}B\bar{C} + ABC = F$

$F = (\bar{A}BC + \bar{A}B\bar{C}) + (A\bar{B}C + \bar{A}\bar{B}C) + ABC$
$= \bar{A}B(C + \bar{C}) + \bar{B}C(A + \bar{A}) + ABC$ ✓
$= \bar{A}B + \bar{B}C + ABC$ ✓
$= \bar{A}B + ABC + \bar{B}C$
$= B(\bar{A} + AC) + \bar{B}C$✓
$= B(\bar{A} + C) + \bar{B}C$✓
$= \bar{A}B + BC + \bar{B}C$✓
$= \bar{A}B + (BC + \bar{B}C)$
$= \bar{A}B + C(B + \bar{B})$✓
$= \bar{A}B + C$✓ (7)

3.2.2 $\overline{\overline{(\bar{A}BC + A\bar{B}C)}\ \overline{(AB\bar{C} + ABC)}} = F$

$F = \bar{A}BC + A\bar{B}C + AB\bar{C} + ABC$ ✓
$= \bar{A}BC + A\bar{B}C + AB(\bar{C} + C)$ ✓
$= \bar{A}BC + A(\bar{B}C + B)$
$= \bar{A}BC + A(C + B)$ ✓
$= \bar{A}BC + AC + AB$ ✓
$= C(\bar{A}B + A) + AB$ ✓
$= C(B + A) + AB$
$= BC + AC + AB$ ✓ (6)

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3.2.3 $\overline{(A+B+\bar{C})(A+\bar{B}+C)(\bar{A}+B+C)(\bar{A}+\bar{B}+\bar{C})} = F$

$F = \overline{(A+B+\bar{C})} + \overline{(A+\bar{B}+C)} + \overline{(\bar{A}+B+C)} + \overline{(\bar{A}+\bar{B}+\bar{C})}$ ✓

$= \bar{A}.\bar{B}.C + \bar{A}.B.\bar{C} + A.\bar{B}.\bar{C} + A.B.C$ ✓

$= \bar{A}(B\bar{C} + \bar{B}C) + A(\overline{B\bar{C}} + BC)$

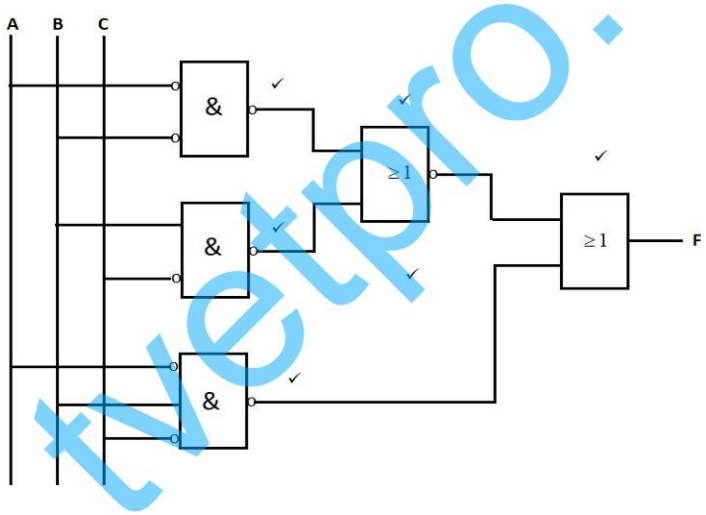
$= \bar{A}(B \oplus C) + A(\overline{B \oplus C})$ ✓

Let $(B \oplus C) = X$ ✓

$\therefore F = \bar{A}.X \oplus A.\bar{X}$ ✓

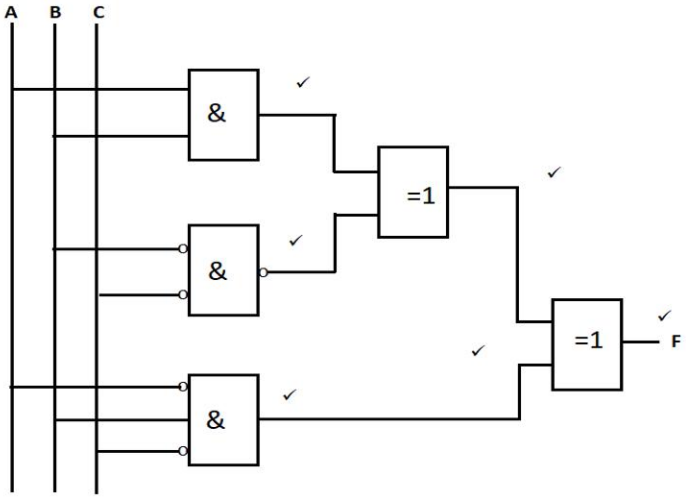
$= A \oplus X$ ✓ (6)

3.3 3.3.1



(6)

3.3.2

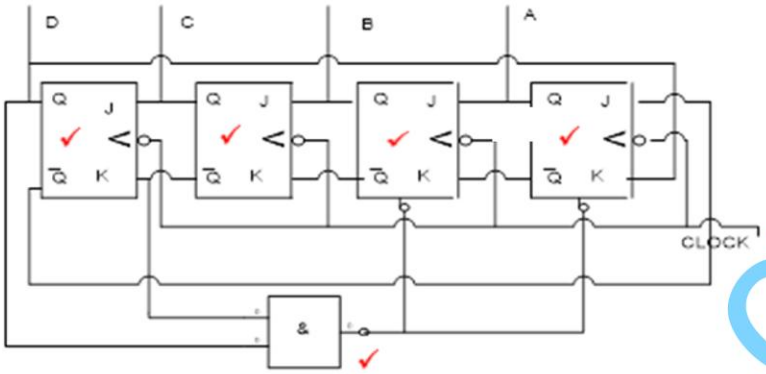


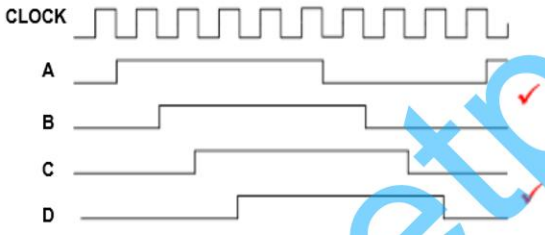
(6)
[34]

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QUESTION 4

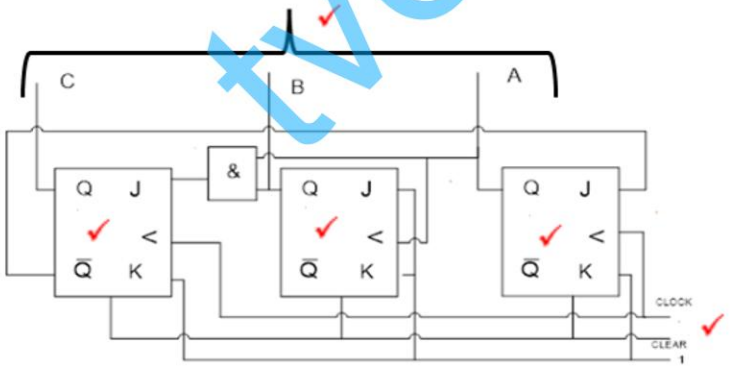
4.1





(8)

4.2



PULSE	C✓	B✓	A✓
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	0	0	0

(8)
[16]

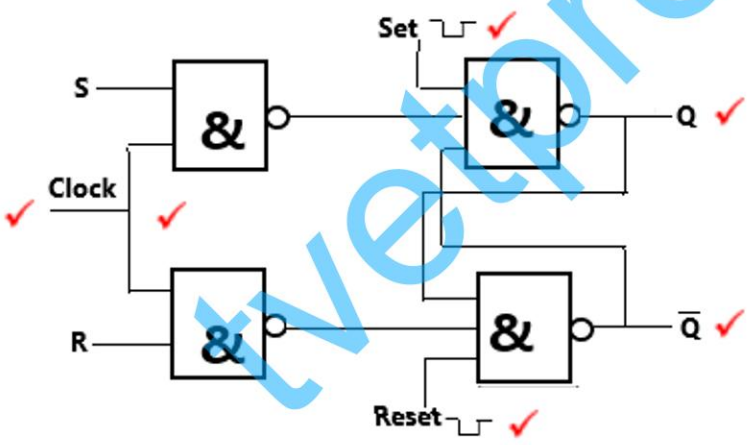
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QUESTION 5

5.1 A decoder converts binary to decimal,✓✓ while an encoder converts decimal to the required code.✓✓ (4)

5.2 The converting process starts with the most significant bit (MSB), then successively tries a 1(one) for every bit.✓ The output of the digital to analogue (D/A) decoder is compared with the input voltage as every bit is compared.✓ When the digital to analogue (D/A) decoder's is greater than the measured voltage (V_m),✓ the '1' in that bit is removed and the next bit (MSB) is compared.✓ If the V_m is greater than reference voltage (V_R), then 1 is kept in that bit and the next bit is compared.✓
After the least significant bit (LSB) has been compared,✓ the digital word in the D/A converter is equivalent of the analogue voltage.✓ (7)

5.3



S	R	Q	$\overline{Q}$	
0	0	Q	$\bar{Q}$	✓
0	1	0	1	✓
1	0	1	0	✓
1	1	*	*	✓

(10)
[21]

TOTAL: 100

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